



Networking and Information Technology Research and Development (NITRD); National Coordination Office (NCO) for/on behalf of the Office of Science and Technology Policy
Subject: AI Action Plan

Utilidata, Inc.

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Angela Kassahun, Director, Policy & Market Development

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan, Federal Register Number 2025-02305

Utilidata appreciates the opportunity to submit comments to the Office of Science and Technology Policy (OSTP) and the National Science Foundation (NSF) in connection with the Administration's Artificial Intelligence (AI) Action Plan.

Utilidata has operated real-time machine learning software on the grid for over a decade. In 2021, Utilidata partnered with NVIDIA to co-develop an AI module custom-designed for the electric distribution grid, called Karman, after recognizing the need for more computing power at the grid edge in order to handle its growing complexity.

Karman is built on an NVIDIA Graphics Processing Unit (GPU) to run distributed AI—the most advanced form of edge computing—allowing electric utilities to run multiple AI models locally without moving vast amounts of sensitive data to a central system. This distributed AI architecture decreases costs and improves reliability, security and outcomes. For example, advanced digital signal processing at the edge combined with local AI models can continuously identify relevant information in real time, such as anomalies that could lead to reliability issues, predict pre-outage conditions or pending equipment failure.

AI-enabled, real-time edge visibility improves grid reliability, identifies outages before they occur and reduces the time it takes to recover from outages. This has a material impact for all customers. Most outage mapping and responses today rely on Advanced Metering Infrastructure (AMI) data, which is dependent on communication system backhaul and more limited data systems, providing information delayed up to one-hour. Deploying distributed AI improves utilities' reliability and resiliency through more granular analysis - sub-minute level - from more sites, more often. It also enhances utilities' ability to more quickly diagnose issues in individual neighborhoods based on correlated grid edge data, such as transformer issues, before infrastructure fully fails.

Finally, distributed AI also makes power more affordable. AI prediction models can be used to forecast customer program impacts and enable real-time program measurement and verification. Utilities can use this capability to inform program enrollment and design, with the result of lower customer energy bills.

As utilities begin to adopt AI technologies, they will require a mix of centralized AI models and distributed AI models that run directly on the infrastructure itself. This architecture is widely utilized across industries, such as automotive, shipping, healthcare, and robotics. Distributed AI is more affordable and secure than solely relying on centralized models that require streaming large amounts of data back to a centralized location. For example, by operating distributed AI in a meter, most of the data would never leave the devices, thereby enhancing security and improving accuracy. These types of architectural decisions will be critical to utilities adopting safe and responsible AI practices.

The electric grid is becoming increasingly complex and traditional physics-based models cannot respond to all of the complexity; instead, our electric grid needs well-managed AI systems to process multiple data sources simultaneously while managing rapid changes in real-time. Utilidata welcomes the opportunity to work with the Administration on AI policy measures that accelerate innovation and adoption of AI-powered technology, like Karman, that improve operations for electric grid infrastructure and provide affordable, reliable, and secure electric power to all Americans.